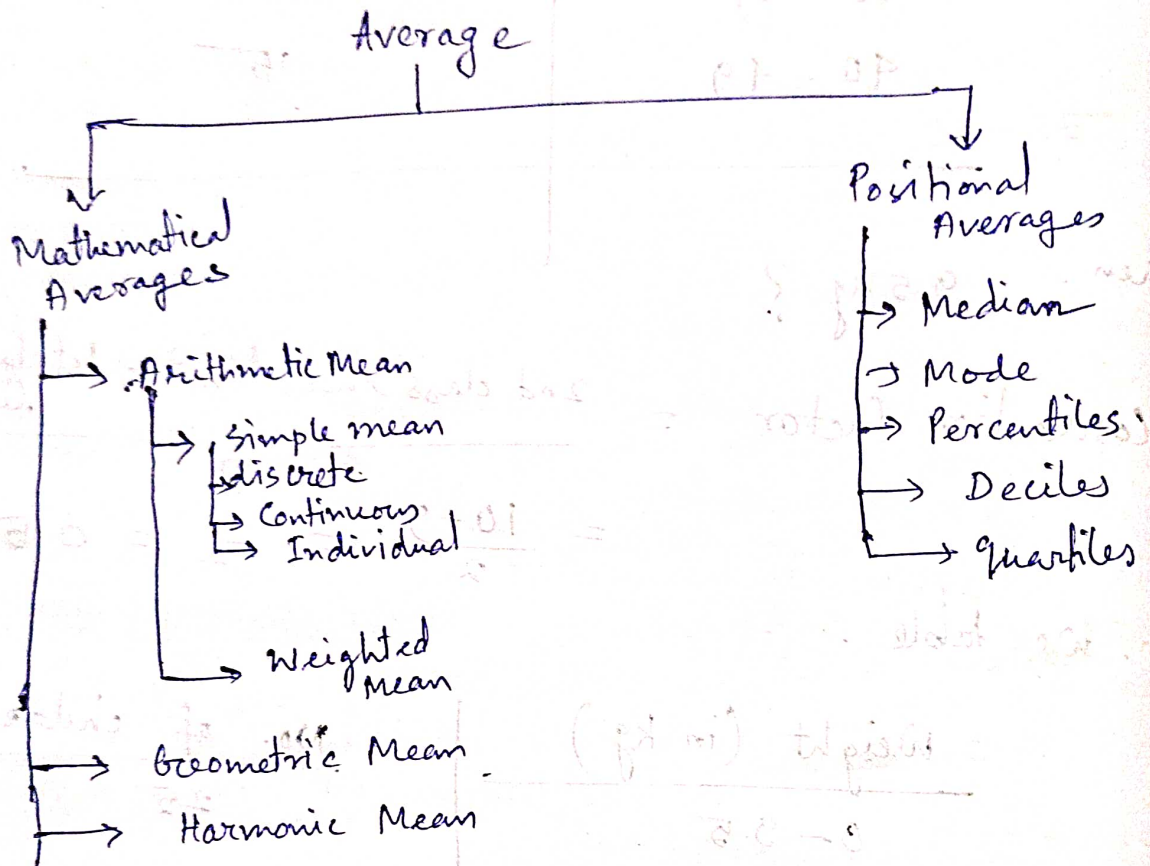


Measure of Central Tendency (Average)

- An attempt ^{Average} to find one single figure to describe whole figures.



Objective of Averages :-

- To get a single value that describe the characteristic of entire group
- To facilitate comparison measure of central value
- It offer the base for computing other measures like variation/ Dispersion, skewness, Kurtosis etc.

Arithmetic Mean:

Simple Mean:

Individual Data $\bar{X} = \frac{X_1 + X_2 + \dots + X_N}{N} = \frac{\sum X}{N}$

Discrete Series $\bar{X} = \frac{\sum fX}{\sum N}$ $X_i = A + \frac{\sum d}{N}$, Here $d = (X - A)$

$$\bar{X} = A + \frac{\sum fd}{N}$$

Continuous Series

$$\bar{X} = \frac{\sum fm}{N}$$

$$\bar{X} = A + \frac{\sum fd}{N} \text{ here } d = (m - A)$$

Individual data:-

Question:- Find the mean of the following data.

11, 29, 17, 21, 13, 31, 39, 19

⇒ The arithmetic mean of the following data is given by

$$\begin{aligned}\bar{X} &= \frac{11 + 29 + 17 + 21 + 13 + 31 + 39 + 19}{8} \\ &= \frac{180}{8} \\ &= 22.5\end{aligned}$$

Arithmetic Mean Calculation by assume mean process.

Given data's are 11, 29, 17, 21, 13, 31, 39, 19

arranging ascending order we have,

11, 13, 17, 19, 21, 29, 31, 39

Now, Let us take assume mean $A = 20$ of the given data.

x	$d = (x - A)$
11	-9
13	-7
17	-3
19	-1
21	1
29	9
31	11
39	19
$\Sigma d = 20$	

$$\therefore \text{A.M. } \bar{X} = A + \frac{\Sigma d}{N}$$

$$= 20 + \frac{20}{8}$$

$$= 20 + 2.5$$

$$= 22.5$$

Ex:- The following data gives the per day production of tablets (in thousand) by a tablet machine 430, 360, 515, 480, 395, 412. Calculate the arithmetic mean for the given data.

Discrete Series

Find the mean of the following:

Mid value	15	20	25	30	35	40	45	50	55
frequency	2	22	19	14	3	4	6	1	1

⇒

Mid value (x)	frequency (f)	f · x
15	2	30
20	22	440
25	19	475
30	14	420
35	3	105
40	4	160
45	6	270
50	1	50
55	1	55

$$\Sigma f = 72$$

$$\Sigma fx = 2005$$

$$\therefore \text{Arithmetic mean } \bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{2005}{72} = 27.847$$

Assumed Mean method

Let us take an assumed mean $A = 30$

Mid value (x)	Frequency (f)	d = (x - A) A = 30	f d
15	2	-15	-30
20	22	-10	-220
25	19	-5	-95
30	14	0	0
35	3	5	15
40	4	10	40
50	1	20	20
55	1	25	25
$\Sigma f = 72$		$\Sigma fd = -155$	

$$\therefore \text{arithmetic mean} = \frac{\Sigma fd}{\Sigma f} = \frac{-155}{72} + 30$$

$$= 30 - 2.152$$

$$= 27.847$$

Ex:- Daily earning (in Rs) of employees working in a Pharmaceutical Company are

Daily Earning (in Rs)	2000	2500	3000	3500	4000
No. of Employees	6	10	22	8	4

Calculate arithmetic mean for given data.

Continuous Series

Calculate the mean of the following:

Size	0-4	4-8	8-12	12-16
frequency	10	12	15	13

⇒

Size(x)	frequency(f)	midpoint(m)	f.m
0-4	10	2	20
4-8	12	6	72
8-12	15	10	150
12-16	13	14	182

$$\Sigma f = 50$$

$$\Sigma(f \cdot m) = 424$$

$$\therefore \text{Mean} = \frac{\Sigma(f \cdot m)}{\Sigma f} = \frac{424}{50} = 8.48$$

By assumed mean method:-

Let us consider to assume mean $A = 10$.

Size	frequency(f)	mid point(m)	$d = (m - A)$	$f \cdot d$
0-4	10	2	-8	-80
4-8	12	6	-4	-48
8-12	15	10	0	0
12-16	13	14	4	52
$\Sigma f = 50$			$\Sigma(f \cdot d) = -76$	

$$\text{Mean} = A + \frac{\Sigma f \cdot d}{\Sigma f} = 10 + \frac{-76}{50} = 10 - 1.52 = 8.48$$

~~Ex. 1~~

Ex. 1

1. Calculate the arithmetic mean of accidents per week for the given data.

No. of Accidents (x)	0-10	10-20	20-30	30-40	40-50
No. of Weeks (f)	10	5	12	9	16

Median

Find the median of the following data
11, 29, 17, 21, 13, 31, 39, 19.

⇒ We arrange the given data in ascending order of magnitude.

11, 13, 17, 19, 21, 31, 39.

Here, the number of observations is 8, which is an even number.

Hence, the median of the data

$$= \frac{1}{2} \left\{ \left(\frac{8}{2} \right)^{\text{th}} \text{ item} + \left(\frac{8}{2} + 1 \right)^{\text{th}} \text{ item} \right\}$$

$$= \frac{1}{2} (4^{\text{th}} \text{ item} + 5^{\text{th}} \text{ item})$$

$$= \frac{1}{2} (19 + 21)$$

$$= 20$$

For an even number

$$\text{Median} = \frac{1}{2} \left\{ \left(\frac{n}{2} \right)^{\text{th}} \text{ item} + \left(\frac{n}{2} + 1 \right)^{\text{th}} \text{ item} \right\}$$

Q. Find the median of 3, 4, 2, 4, 4, 1, 5.

⇒ We arrange the given data in ascending order
1, 2, 3, 4, 4, 4, 5
 ↓
 4th item.

∴ Here, the number of observation is 7 which is an odd number,

Hence, the median of the data,

$$= \frac{(n+1)}{2} \text{th item}$$

$$= \frac{7+1}{2} \text{th item}$$

$$= 4 \text{th item}$$

$$= 4.$$

Q. Find the median of the following.

Variable	15	20	25	30	35	40	45	50	55
frequency	2	22	19	14	3	4	6	1	1

⇒

Variable (x)	frequency (f)	Cummulative frequency (Cf)
15	2	2
20	22	24
<u>25</u>	19	43
30	14	57
35	3	60
40	4	64
45	6	70
50	1	71
55	1	<u>72</u>

$$\frac{72+1}{2} = \frac{73}{2} = 36.5^{\text{th}} \text{ value} = 25$$

Q. Calculate the median of the following

Size	0-4	4-8	8-12	12-16
frequency	10	12	15	13

⇒

Size	frequency	Cummulative frequency
0-4	10	10
4-8	12	22
8-12	15	37
12-16	13	50

$$\frac{50+1}{2} = 25.5 \text{ th value.}$$

$$\text{Median} = l_1 + \left(\frac{\frac{N}{2} - c.f}{f} \right) \times i$$

$$= 8 + \frac{\frac{50}{2} - 22}{15} \times 4$$

$$= 8 + \frac{3}{15} \times 4$$

$$= 8 + 0.8$$

$$= 8.8$$

Ex. Find median for the following distribution

Variable	0-20	20-50	50-100	100-250	250-500	500-1000
Frequency	20	50	80	30	20	10

Mode

- Mode is the most frequent value of the series

Example (Individual Series)

Find the mode of the numbers

3, 2, 5, 4, 4, 2, 4, 3, 3, 4, 4, 5, 4, 2, 4, 4, 2, 4, 5, 4, 4

⇒ The frequency distribution of the given set of numbers is

Variable	2	3	4	5
frequency	4	3	11	3

We see that the variable value 4 has occurred most frequently, that is, has maximum frequency. Therefore, the required mode of the given number is 4.

Example (Discrete Series)

Find the mode of the following

Variable	4	6	8	10	12	16
frequency	1	2	2	1	1	1

⇒ We see that the variable values 6 and 8 have occurred most frequently, that is, have maximum frequency 2. Therefore, the given set of observations possesses two modes viz, 6 and 8.

Example (Continuous Series)

Calculate the mode value of the given data.

Variable	0-10	10-20	20-30	30-40	40-50
frequency	10	5	16	9	12

⇒ Given data $L = 20$, $f_1 = 16$, $f_0 = 5$, $f_2 = 9$
 $i = 10$

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

$$= 20 + \frac{16 - 5}{(2 \times 16) - 5 - 9} \times 10$$

$$= 20 + \frac{11}{32 - 14} \times 10$$

$$= 20 + \frac{11}{18} \times 10$$

$$= 20 + 6.1 = 26.1$$

∴ Mod value = 26.1

Ex:- Calculate the mode value of the following

Size	0-4	4-8	8-12	12-16
frequency	10	12	15	13

Relationship between mean, median and mode:

$$\boxed{\text{Mode} = 3 \text{ median} - 2 \text{ mean}}$$

Example:- For a moderately asymmetric distribution, median = 27, mean = 26. Find mode.

⇒ For a moderately asymmetric distribution

$$\begin{aligned}\text{Mode} &= 3 \text{ median} - 2 \text{ mean} \\ &= 3 \times 27 - 2 \times 26 \\ &= 81 - 52 \\ &= 29\end{aligned}$$

Ex 1:- If the mean and median of a frequency distribution are 20 and 17 respectively. find the mode.

Ex 2:- If Median of the series is given as 32 and mode is 34. find the mean of the data.

Ex 3:- In a distribution, the mode and mean are 25 and 28 respectively. Find out the value of Median.